

# Energy Shocks and the Green Adoption Channel according to E-HANK

Model, experiments, and caveats

Guillaume, Boris Chafwehe, Francesca Diluiso

July 24, 2026

## Abstract

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of [Auclert et al. \(2024\)](#). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks, and the two fiscal responses studied in the literature: a retail price cap ([Langot et al., 2026](#)) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption ([Bayer et al., 2026](#)). Our central result is an interaction neither paper can see: **the price cap stabilises output but mechanically eliminates the green adoption margin**, while the transfer delivers comparable stabilisation and leaves adoption fully intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. The loss is linear in the cap’s intensity under a price shock and convex under a supply shock, and the *sign* of the cap’s welfare effect flips with the elasticity of energy supply — reconciling [Langot et al. \(2026\)](#) and [Bayer et al. \(2026\)](#) within a single model.

## 1 Contribution

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. [Auclert et al. \(2024\)](#) study a small open economy and stress the negative externality when all importers subsidise simultaneously. [Bayer et al. \(2026\)](#) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. [Langot et al. \(2026\)](#) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well.

All three treat the household’s energy technology as fixed. We relax exactly that. The green durable is an extensive margin: the energy shock raises the return to adoption, so the crisis itself accelerates the transition — unless policy removes the price signal that drives it.

Our energy-market closure nests the disagreement between [Bayer et al. \(2026\)](#) and [Langot et al. \(2026\)](#) in a single parameter: with  $\varepsilon^E = \infty$  the economy is a price taker (elastic supply, Langot); with  $\varepsilon^E$  finite the quantity is fixed and the world price clears the market (Bayer). Both are reported below.

## 2 Model

### 2.1 Environment

The aggregate structure is [Auclert et al. \(2024\)](#): a small open economy with sticky wages, a CES consumption basket nesting energy against a home/foreign bundle, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share  $\zeta^E$  of the energy endowment. The numeraire is the CPI basket, so all relative prices carry a  $P$  subscript.

## 2.2 The durable state

Each household carries a durable state  $d$ , the *pair* (holding entering the period, holding chosen last period), needed to charge the switching cost:

$$d \in \{BB, BG, GB, GG\},$$

where  $GB$  (green now, brown before) is the state that pays  $\psi_g$ . Brown is absorbing; green breaks down to brown at rate  $\delta_g$ . The state  $BG$  carries zero mass by construction, so the effective state is three-point.

Stages, in forward order, are [**dep**, **prod**, **durables**, **consav**]: breakdown, idiosyncratic productivity, the discrete adoption choice (a logit with taste scale  $\sigma_\varepsilon$ ), and the EGM consumption–savings step.

## 2.3 How the durable enters the budget

This is the substantive modelling decision. Households of different durable types face different energy prices, so they face different *consumption bundle prices*. We give each type its own exact CES cost-of-living index:

$$p_d^E = p_{hh}^E + \mathbb{I}[d \text{ green}] (p_g^E - p_{hh}^E), \quad p_d = \left[ 1 + \alpha_E ((p_d^E)^{1-\eta_E} - (p_{hh}^E)^{1-\eta_E}) \right]^{\frac{1}{1-\eta_E}},$$

and the budget constraint becomes  $p_d c + a' = \text{coh}(d)$ , with  $\text{coh}$  carrying only the switching cost. The second expression uses the CES price identity enforced by the equilibrium block, so brown households have  $p_d \equiv 1$  *exactly*, at every date and with structurally zero derivatives. This matters numerically: writing  $p_d$  in the naive form leaves the identity to be recovered by composing Jacobians along the DAG, which left  $\sim 7 \times 10^{-6}$  of residual and broke the nesting test.

**What this replaces.** An earlier version booked the energy price gap as a first-order income transfer with energy quantity frozen at its steady state. That is valid to first order but shuts down the intensive margin by construction: adopters could not re-optimize energy use. The CES formulation above is exact and lets adopters respond to their own price.

**Aggregation.** With type-specific prices, the aggregate energy demand is  $\sum_d c_E(d)$ , *not* the CES demand evaluated at the aggregate index — these differ by Jensen’s inequality (0.5% here). Aggregate  $c_H$ ,  $c_F$ ,  $c_E$  are therefore built from the household block’s own hetoutputs. Getting this wrong breaks asset market clearing by 0.002–0.06.

## 2.4 Balance of payments

Two durable-margin terms enter the current account.

- **Switching cost as an import**,  $\text{imports}_{dur} = \psi_g D^{sw}$ . Booking it as domestic demand fails here: with  $\mu_{ss} > 1$  and capitalised profits, extra demand for the home good creates a marginal dividend owned by nobody and breaks asset clearing by its capitalised value.
- **Energy gap as a real import saving**,  $(p_{hh}^E - p_g^E) C_E^G$ . *This is a modelling choice, not an accounting patch*: it asserts that adopters’ cheaper energy is a genuine resource saving for the country, not a domestic transfer. Without it the windfall has no external counterpart and asset clearing fails by 0.5–0.7.

## 2.5 Policy instruments

Both instruments are the extreme (full-compensation) versions, as in [Bayer et al. \(2026\)](#).

$$\text{Price cap: } p_{hh,t}^E = (1 - \tau^E)p_t^E + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow \text{retail price fixed at pre-crisis level} \quad (1)$$

$$\text{Slutsky transfer: } tr_{it} = \iota (p_t^E - \bar{p}^E) \bar{c}_{E,i}, \quad \iota = 1 \Rightarrow \text{full compensation on } pre\text{-crisis use} \quad (2)$$

Both are debt financed and repaid through the distortionary labour tax. The transfer base  $\bar{c}_{E,i}$  is each household's steady-state energy demand; we verified it aggregates to the government's cost base to machine precision.

**Why the two differ for adoption.** The cap operates on  $p_{hh}^E$ , which is exactly the price entering the brown–green gap. It therefore compresses the adoption incentive directly. The transfer is lump-sum with respect to current choices and leaves the relative price intact.

## 3 Calibration

| Parameter                            | Value       | Source / target                       |
|--------------------------------------|-------------|---------------------------------------|
| $\alpha_E$ energy expenditure share  | 0.04        | <a href="#">Auclert et al. (2024)</a> |
| $\eta_E$ energy substitution         | 0.10        | <a href="#">Auclert et al. (2024)</a> |
| $r, \mu_{ss}$                        | 0.01, 1.03  | <a href="#">Auclert et al. (2024)</a> |
| $\beta$ heterogeneity (3 types)      | spread 0.06 | MPC $\approx$ 0.34                    |
| $\delta_g$ green breakdown rate      | 0.05        | assumption (5 yr life)                |
| $p_g^E/p^E$ steady-state gap         | 0.80        | assumption                            |
| $\psi_g$ switching cost              | 0.253       | targets $D_{ss}^G = 0.05$             |
| $\sigma_\epsilon$ taste scale        | $10^{-2}$   | <b>not identified (§5)</b>            |
| $\rho_g$ pass-through to green price | 0           | <b>extreme case (§5)</b>              |

Table 1: Calibration. The steady state is identical across all experiments.

## 4 Experiments and results

**Shocks.** The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life (ARS). The *supply* shock is a 20% cut in available energy for 6 quarters ([Bayer et al., 2026](#)); the world price then clears the market endogenously.

| Shock  | Policy   | $y(0)\%$ | $\sum y$ | $\pi(0)$ ann. | peak $D^G\%$ | fiscal cost | adoption $\rightarrow \sum y$ |
|--------|----------|----------|----------|---------------|--------------|-------------|-------------------------------|
| price  | none     | −0.997   | −23.9    | 9.47          | <b>31.86</b> | 0.00        | +4.32                         |
| price  | cap      | −0.024   | −21.3    | −1.56         | −0.01        | 53.96       | −0.04                         |
| price  | transfer | +1.322   | +18.7    | 11.40         | <b>31.92</b> | 53.52       | +4.35                         |
| supply | none     | −1.368   | −15.6    | 21.26         | <b>20.77</b> | 0.00        | +1.38                         |
| supply | cap      | −1.515   | −94.0    | −1.91         | −0.07        | 52.17       | +0.10                         |
| supply | transfer | +3.064   | +9.5     | 31.13         | <b>30.37</b> | 60.91       | +8.84                         |

Table 2: Summary. Cumulative sums over 24 quarters, in % deviations. The last column is the contribution of the adoption margin, computed as the difference between the model with the margin open and the same model with it shut.

#### 4.1 Result 1: the crisis accelerates the transition

With no policy, the green share rises by up to 32 log points (price shock) and 21 (supply shock). The adoption margin is expansionary for cumulative output (+4.3 and +1.4), because adopters escape the energy bill and their real income falls less.

#### 4.2 Result 2: the price cap eliminates the adoption margin

This is the paper’s headline. Under the cap, the peak green share response falls from +31.9 to −0.01 log points under the price shock, and from +20.8 to −0.07 under the supply shock. **The adoption channel is not attenuated; it is switched off.** Its contribution to cumulative output collapses from +4.32 to −0.04. The mechanism is transparent: the cap holds  $p_{hh}^E$  at its pre-crisis level, and  $p_{hh}^E$  is precisely what determines the return to adoption.

#### 4.3 Result 3: the transfer stabilises without this cost

The Slutsky transfer leaves the green share response essentially unchanged (+31.9 vs +31.9 under the price shock) while delivering more output stabilisation than the cap, at similar fiscal cost. This strengthens Bayer et al. (2026)’s case for transfers over subsidies with an argument they do not make: subsidies also destroy the transition margin.

#### 4.4 Result 4: with fixed supply, the cap is destructive

Under the supply shock the cap yields cumulative output of −94 against −15.6 with no policy. This replicates Bayer et al. (2026)’s core finding — with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief — and shows it survives the addition of an adoption margin. It is also precisely the configuration in which Langot et al. (2026), assuming elastic supply, find the cap effective. Our elastic-supply column agrees with them (−21.3 vs −23.9, a mild improvement).

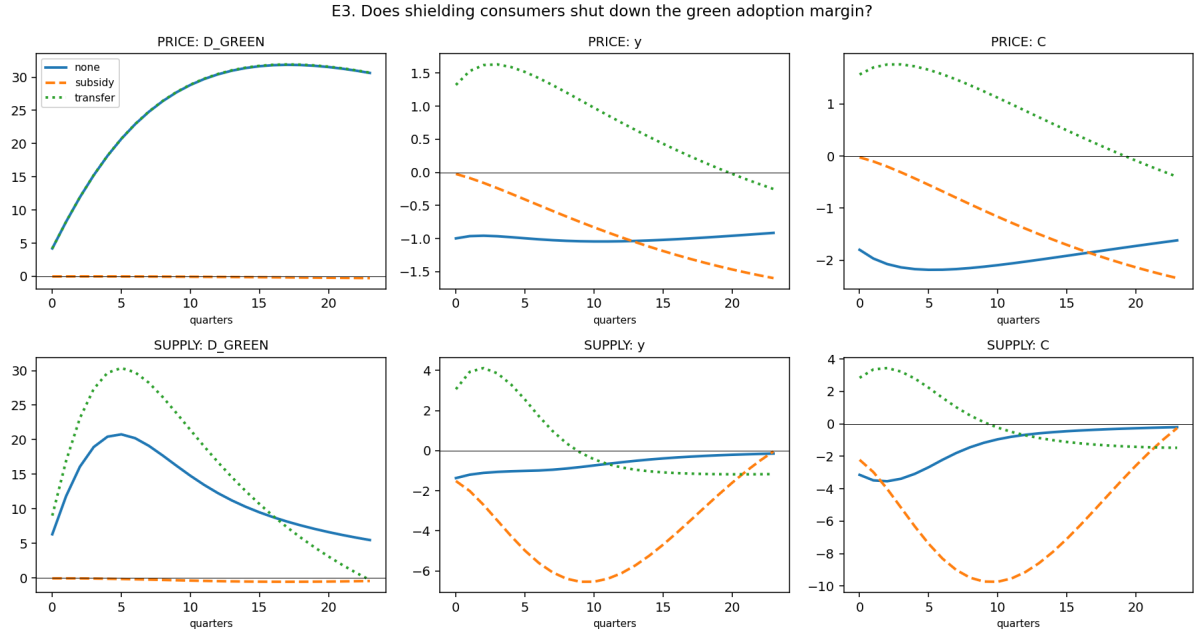


Figure 1: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

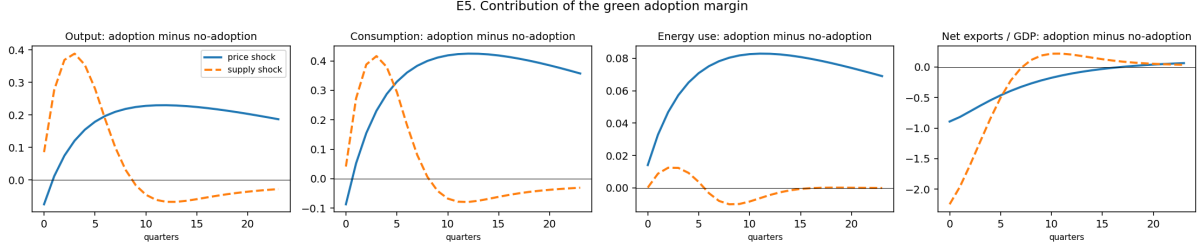


Figure 2: Contribution of the adoption margin: difference between the model with the margin open and with it shut.

#### 4.5 Result 5: the cap is a dial, not a switch

Section 4.2 used  $\tau^E = 1$ , a complete price freeze. No government did that. Table 3 sweeps  $\tau^E \in [0, 1]$ .

| Shock  | Policy              | peak $D^G$ | $\sum y$ | $\pi(0)$ | fiscal | CEV %  |
|--------|---------------------|------------|----------|----------|--------|--------|
| price  | cap $\tau^E = 0$    | 31.86      | -23.9    | 9.47     | 0.00   | -2.008 |
| price  | cap $\tau^E = 0.25$ | 23.88      | -23.2    | 6.74     | 13.43  | -1.901 |
| price  | cap $\tau^E = 0.50$ | 15.88      | -22.6    | 3.99     | 26.90  | -1.794 |
| price  | cap $\tau^E = 0.75$ | 7.87       | -21.9    | 1.23     | 40.41  | -1.685 |
| price  | cap $\tau^E = 1$    | -0.01      | -21.3    | -1.56    | 53.96  | -1.575 |
| price  | transfer            | 31.92      | +18.7    | 11.40    | 53.52  | +0.129 |
| supply | cap $\tau^E = 0$    | 20.77      | -15.6    | 21.26    | 0.00   | -1.712 |
| supply | cap $\tau^E = 0.25$ | 19.55      | -21.0    | 19.50    | 13.11  | -1.944 |
| supply | cap $\tau^E = 0.50$ | 17.43      | -30.1    | 16.86    | 32.99  | -2.307 |
| supply | cap $\tau^E = 0.75$ | 13.12      | -47.5    | 12.22    | 63.57  | -2.960 |
| supply | cap $\tau^E = 1$    | -0.07      | -94.0    | -1.91    | 52.17  | -4.684 |
| supply | transfer            | 30.37      | +9.5     | 31.13    | 60.91  | +0.179 |

Table 3: Dose-response in the cap intensity. CEV is the consumption-equivalent variation of the whole scenario relative to the steady state; negative means households would pay that share of permanent consumption to avoid it.

Two different shapes. Under the *price* shock the adoption loss is almost exactly **linear** in  $\tau^E$ : half a cap costs half the transition (15.88 against 31.86). Under the *supply* shock it is **convex**: small caps barely touch adoption (19.55 at  $\tau^E = 0.25$ ) and the collapse is concentrated near the full freeze. A partial French-style shield is therefore much less damaging to the transition when energy supply is inelastic — but, as the next result shows, much more damaging to everything else.

#### 4.6 Result 6: the sign of the welfare effect flips with the supply elasticity

This is the sharpest result in the paper. Under the price shock, tightening the cap *raises* welfare monotonically, from -2.01% to -1.58%. Under the supply shock it *lowers* welfare monotonically, from -1.71% to -4.68% — almost tripling the cost of the crisis.

Langot et al. (2026) assume elastic supply and conclude the French shield worked. Bayer et al. (2026) assume fixed supply and conclude subsidies are destructive. **Both are right within their own closure, and our model reproduces each by moving a single parameter.** The disagreement in the literature is not about mechanisms; it is an empirical question about the elasticity of energy supply at the relevant horizon.

## 4.7 Result 7: distributional incidence, and its reversal

| Scenario (price shock) | CEV mean | impatient | middle | patient |
|------------------------|----------|-----------|--------|---------|
| no policy              | −2.008   | −2.944    | −2.406 | −0.674  |
| cap $\tau^E = 0.5$     | −1.794   | −2.255    | −2.145 | −0.980  |
| cap $\tau^E = 1$       | −1.575   | −1.556    | −1.882 | −1.288  |
| transfer               | +0.129   | +0.482    | −0.012 | −0.083  |

Table 4: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs.

Without policy the impatient lose four times as much as the patient (−2.94 vs −0.67). The full cap **reverses the incidence**: the impatient now lose *less* than the patient (−1.56 vs −1.29), because the cap is financed by future distortionary taxes falling on everyone while its benefit is concentrated on those who cannot smooth. This reproduces Bayer et al. (2026)’s finding that the subsidy shifts the burden from the poor to the rich, in a model with a different heterogeneity mechanism — theirs comes from energy-share heterogeneity, ours from discount-factor heterogeneity alone.

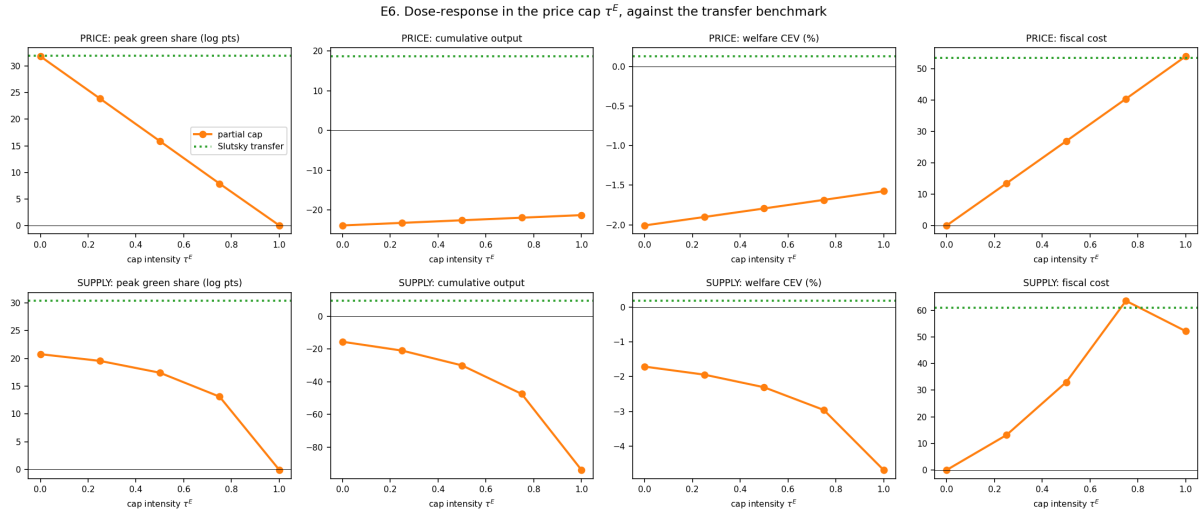


Figure 3: Dose-response in  $\tau^E$  (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

## 5 Caveats

We list these because several are load-bearing.

1.  **$\psi_g$  and  $\sigma_\varepsilon$  are not separately identified.** A single target  $D_{ss}^G = 0.05$  is hit by a continuum of pairs, and they imply very different adoption *elasticities* — which is the object every counterfactual depends on. At  $\psi_g = 0.20$ , moving  $\sigma_\varepsilon$  from  $10^{-2}$  to  $3 \times 10^{-3}$  moves  $D^G$  by three orders of magnitude.  $D^{sw}$  cannot serve as a second target: it satisfies  $D^{sw} = \delta_g D^G$  mechanically. *An empirical semi-elasticity of adoption to a price differential or subsidy is required.* Until then the magnitudes in Table 2 should be read as illustrative.
2. **Zero pass-through to the green price.**  $\rho_g = 0$  means adopters are *completely* insulated from the fossil shock. This is the upper bound on the adoption channel. A partial pass-

through calibrated to observed electricity–gas co-movement would be more defensible; the machinery is in place ( $\rho_g$  is a parameter), only the calibration is missing.

3. **The transfer is too expansionary, and its CEV is not credible.** Output rises +1.3% (price) and +3.1% (supply) on impact, and the welfare CEV is *positive* (+0.13%, +0.18%). A positive CEV means households are better off *with* the energy crisis plus the transfer than in the steady state without either. That cannot be right. We verified the transfer aggregates to the government’s cost base to machine precision, so it is not an accounting leak; it reflects full Slutsky compensation combined with MPC  $\approx 0.34$ , debt financing with slow repayment ( $\psi_B = 0.04$ ), and critically **the absence of an investment margin** — Bayer et al. (2026) have capital, which crowds out. *The transfer column of Table 3 should be treated as an upper bound, not an estimate.* The *ranking* of transfer over cap on the adoption margin is robust, since it follows from the relative price alone; the *level* of the transfer’s welfare gain is not.
4. **The supply shock produces a very large price response** (+371 log points for a 20% quantity cut, against  $\sim 175$  in Bayer et al. (2026)). With  $\eta_E = 0.1$  and energy entering consumption only (prod<sub>E</sub> share = 0), demand is extremely inelastic, so the price must do all the work. Adding energy to production would damp this considerably. **This number should not go into a draft as it stands.**
5. **Linearity.** All results are first-order. A 20% energy cut is not marginal; Bayer et al. (2026) report that second-order solutions amplify but do not reshape the responses.
6. **The numeraire.** We implemented and tested a domestic-good numeraire. The steady state is bit-identical and real IRFs agree to  $\sim 4 \times 10^{-4}$ , confirming it is a relabeling. A residual mismatch of about 4% of the IRF magnitude could not be localised: it is invariant to the grid size ( $n_a$  from 150 to 600) and to the grid bound (max  $a$  from 400 to 4000), so it is not discretisation. Because the CPI numeraire is used throughout, this does not affect any result reported here. *Reported for completeness, unresolved.*
7. **No within-country energy-share heterogeneity.** Bayer et al. (2026)’s central distributional mechanism is that poor households have higher energy shares. We have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.

## 6 Premortem: how Boris would attack this

1. “Your headline result is mechanical.” The cap fixes  $p_{hh}^E$ ; adoption depends on  $p_{hh}^E$ ; therefore adoption dies. This is an identity, not a finding. **Response:** the mechanism is transparent, but the *magnitude* is not — whether the transition loss is second-order or first-order relative to the stabilisation gain is quantitative, and it depends on the adoption elasticity we have not yet identified. The honest framing is that the paper quantifies a trade-off, not that it discovers one. This makes caveat (1) existential rather than cosmetic.
2. “A real cap is not 100%.” **Answered** in Section 4.5: the dose-response is linear under the price shock and convex under the supply shock. The remaining version of the objection is that the French shield capped the *growth rate* of the regulated tariff, not its level, which is a different instrument again.
3. “Your welfare numbers say the crisis makes people better off.” The transfer’s CEV is positive. This is the clearest symptom that the fiscal block is under-disciplined: no investment crowding out, slow debt repayment, and full compensation. **Response:** the welfare *ordering* across policies is informative; the levels are not. This should be fixed before the welfare section is written up, not defended.

4. “*Why is the green price exogenous?*” If everyone adopts, green energy demand rises and its price should rise too. With  $\rho_g = 0$  and no green supply curve, adoption has no general-equilibrium cost, which biases the channel upward.
5. “*You have no investment and no energy in production.*” Both [Bayer et al. \(2026\)](#) and [Auclert et al. \(2024\)](#) have richer supply sides. Ours makes the transfer look too expansionary and the supply shock too inflationary. A referee will notice the 371 log-point price response.
6. “*The green durable is not a durable.*” It has no stock, no resale, no maintenance — only a switching cost and a breakdown hazard. It is a technology-adoption indicator. Calling it a durable invites comparison to a literature we do not engage.
7. “*The energy-gap block is doing a lot of work.*” Booking adopters’ saving as a national import saving is what makes the channel expansionary. The alternative — a domestic transfer financed by someone — would change the aggregate result materially. This choice should be defended in the text, not in a docstring.
8. “*Welfare is missing.*” All three reference papers compute consumption-equivalent variation. Without it we cannot say the transfer is *better*, only that it stabilises output while preserving adoption.

## 7 Replication

|                      |                                                    |
|----------------------|----------------------------------------------------|
| ehank/calibration.py | BASE / DURABLE / POLICY layers; make_calibration() |
| ehank/blocks.py      | aggregate blocks, [ARS] vs [NEW] marked            |
| ehank/household.py   | durable household (StageBlock + LogitChoice + EGM) |
| ehank/model.py       | assembly, steady state, shocks, run()              |
| run_experiments.py   | full matrix -> output/                             |

A run is fully described by its overrides: `run(m, shock_kind='supply', policy='subsidy', model_variant='adoption')`. The steady state is identical across all experiments. `test_targets()` is called after every solve because `solve_steady_state` does not check its own residuals.

## References

- Auclert, A., H. Monnery, M. Rognlie, and L. Straub (2024). “Managing an Energy Shock: Fiscal and Monetary Policy.” In *Heterogeneity in Macroeconomics: Implications for Monetary Policy*, Series on Central Banking, Analysis, and Economic Policies, Vol. 30, Central Bank of Chile, 39–108.
- Bayer, C., A. Kriwoluzky, G. J. Müller, and F. Seyrich (2026). “Redistribution Within and Across Borders: The Fiscal Response to an Energy Shock.” *Journal of Political Economy Macroeconomics*, forthcoming. Earlier version: “Hicks in HANK,” CEPR DP 18557.
- Langot, F., S. Malmberg, F. Tripier, and J.-O. Hairault (2026). “The Macroeconomic and Redistributive Effects of the French Tariff Shield.” *Journal of Political Economy Macroeconomics*, forthcoming. Earlier version: CEPREMAP Working Paper 2305.